



Polyethylene glycol/graphene oxide aerogel shape-stabilized phase change materials for photo-to-thermal energy conversion and storage via tuning the oxidation degree of graphene oxide

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ARTICLE INFO

Article history:

Received 18 March 2017
Received in revised form 13 May 2017
Accepted 15 May 2017
Available online 23 May 2017

Keywords:

Oxidation level
Graphene oxide aerogel
Composite phase change materials
Shape-stabilization
Photo-to-thermal energy conversion and storage

ABSTRACT

Polyethylene glycol (PEG)/graphene oxide aerogel (GA) composite phase change materials (PCMs) were prepared by introducing PEG into GAs from graphene oxide (GO) with different oxidation degree via vacuum impregnation. The structures of GAs were tuned by the oxidation levels of GO. A series of characterizations were used to analyze the chemical structure of GOs, including X-ray diffraction (XRD), Fourier transform infrared spectrum (FTIR), X-ray photoelectron spectroscopy (XPS) and Raman spectroscopy analysis. Structural analyses confirmed that the oxygenated functional groups increased and the hydroxyl groups were transformed into carboxyl and epoxy groups with increasing oxidation level. In addition, the graphitic nature of GO decreased while the sp^3 domains of GOs increased owing to the disruption of the graphitic stacking order. Morphology analysis showed that the breakage of graphene sheet became more serious with the oxidation level increasing. When GAs prepared with GOs of higher oxidation levels were used, the composite PCMs showed excellent shape-stability during phase change and excellent thermal repeatability. The change of dimension for PGA6-40 heated from 35 °C to 150 °C was negligible under the load of a constant force (7 N). Efficient photo-to-thermal energy conversion and storage was realized in the composite PCMs.

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1. Introduction

The sustainable development in energy is of critical importance for human beings because a great amount of energy available is limited, intermittent and unbalanced in nature. The increase of greenhouse gases and the shortage of fossil fuels also drive the development and efficient utilization of various kinds of renewable energy. Therefore, energy storage plays a crucial role in saving energy and alleviating the contradiction between the demand and the supply of energy. Thermal energy storage (TES), one popular method of energy storage, can be achieved by sensible heat thermal energy storage, latent heat thermal energy storage and the chemical energy storage of the reversible chemical reaction. Among them, the applications of TES are based on the latent heat thermal energy storage (LHTES) of phase change materials (PCMs) at present, and PCMs have become an integral part of the LHTES system with high energy storage density and isothermal characteristics during heat storing and releasing [1–5].

During melting or solidification, PCMs can store or release a large amount of energy at a nearly constant temperature [6]. As advanced energy-saving materials with so many advantages, PCMs have exhibited important application value and broad development prospects in many fields, including thermal energy management system [3,5,7], aerospace, solar energy utilization [8–11], textile industry [12,13], thermal storage building [14–16], etc. [17,18].

There are many substances that can be used as PCMs, which can be divided into four categories according to the phase change process: solid–solid PCMs, solid–liquid PCMs, solid–gas PCMs and liquid–gas PCMs. Nowadays, solid–liquid PCMs are more widely studied and applied. Solid–liquid PCMs exhibit a solid–liquid transition during phase change, and the liquid phases show a certain fluidity which restricts their applications. Shape stabilization of the PCMs has been proved to be a solution. Shape-stabilized PCMs comprised of working medium and supporting materials have been widely explored, in which the supporting structures and materials can prevent the working medium from leaking during phase transition since 1990s [19]. Many microencapsulation methods of PCMs have been developed from polymer shells to graphite based materials (expanded graphite (EG) [11,20], graphene oxide (GO)

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